AKANKSHA SHARMA

B.S. in AI and Data Science

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EDUCATION

Indian Institute of Technology (IIT) Jodhpur

First-Year Undergraduate Student | B.S. in AI and Data Science

TECHNICAL & SOFT SKILLS

Languages & Core Tech: Python, JavaScript, CSS, HTML
AI, Data Science & Quantum: Artificial Intelligence, Large Language Models (Ollama, OpenAI API), RAG Pipelines, Qiskit, Fujitsu QARP, Machine Learning
Tools & Libraries: Streamlit, FastAPI, Docker, PostgreSQL (pgvector), NumPy, SciPy, yfinance, QGIS, Git
Soft Skills & Leadership: Public Speaking, Extempore & Debate, Social Advocacy, Cross-functional Collaboration, Sustainable Development

COLLEGE PROJECTS & EXPERIENCE

Quantum-Alpha Portfolio Optimizer

Tech Stack: Python, VS Code, Streamlit, FastAPI, Qiskit, yfinance

- Developed a full-stack local web application using Streamlit and FastAPI, integrating yfinance market data with a Qiskit quantum backend to deliver interactive, chatbot-driven portfolio risk analysis and asset optimization.

Bitmark Hackathon 2026 Participant

Tech Stack: Python, OpenAI API, Bitmark Standard, HTML/CSS, Git

- Built an AI-powered editor leveraging the Bitmark open standard to transform natural language prompts into interactive educational content within a high-pressure, 24-hour hackathon environment.
- Focused on rapid prototyping and user-centric design to automate structural content generation, earning recognition for usability and technical innovation.

Hiring Intelligence System (AI Founder - OS)

Tech Stack: Python, Streamlit, Ollama, Docker, PostgreSQL (pgvector)

- Architected and deployed a locally hosted Hiring Intelligence System utilizing a Retrieval-Augmented Generation (RAG) pipeline to streamline candidate evaluation.

Indian Space Academy (ISA)

Internship & Training

- Completed specialized training and internship focused on Remote Sensing and applied geospatial analysis using QGIS.
- Mapped and analyzed changes in forest cover and urban industrialization over a period of time to track ecological and infrastructural developments.

Quantum Finance Research Project

Ongoing | Tech Stack: Python, Fujitsu QARP, Qiskit, NumPy, SciPy

- Formulated portfolio risk optimization problems into Quadratic Unconstrained Binary Optimization (QUBO) models to execute on the Fujitsu Quantum-Inspired Analysis Radio Protocol (QARP).
- Designed and simulated Quantum Approximate Optimization Algorithm (QAOA) circuits, benchmarking scalable quantum approaches against classical Markowitz portfolio optimization methods.

SCHOOL PROJECTS & INNOVATIONS

Smart Traffic Diversion System

Exhibited at Young Creators League, Plaksha University | Selected for National Level Exhibition

- Designed and prototyped an intelligent traffic management system utilizing sensor arrays and camera-based speed detection to monitor real-time traffic intensity.
- Created a backend logic controller to dynamically prompt specific traffic lights, effectively diverting vehicles and reducing congestion.

Blind-Curve Accident Alert System

Selected for National Level Science Exhibition (Won UT Level)

- Engineered a proactive, sensor-based vehicle alert system designed to detect oncoming traffic at blind curves, mitigating the risk of head-on collisions.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Girls India Project & CCPCR Advocate

Chandigarh Commission for Protection of Child Rights (CCPCR)

- Participated actively as a member and advocate in Mock Parliaments, engaging in policy discussions and youth advocacy.

Community & Communication Initiatives

- **Public Speaking & Debate:** Active participant and competitor in Model United Nations (MUNs), parliamentary debates, extempores, and science quizzes.
- **Sustainability & Social Impact:** Planted 105 saplings leading environmental conservation efforts, and leveraged theater club performances to spread critical social messages.